Outlier Detection

Statistical, proximity-based, clustering-based and classification-based approaches

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Source: J. Han, M. Kamber & J. Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann, 2011 — Chapter 12

Roadmap

1. Outliers and outlier analysis
2. A taxonomy of detection methods
3. Statistical approaches
4. Proximity-based approaches
5. Clustering-based approaches
6. Classification-based approaches
7. Contextual and collective outliers
8. High-dimensional data
9. Practice and summary

Learning outcomes

- Distinguish outliers from noise; classify them as global, contextual or collective
- Apply the 3σ rule, Grubbs' test and the Mahalanobis distance
- Compute DB(r, π) outliers and LOF by hand
- Use clustering- and classification-based detectors
- Adapt detection to context, structure and high dimensionality

Outliers and Outlier Analysis

Section 1

What is an outlier?

> **Definition** An **outlier** is a data object that deviates significantly from the rest of the objects, *as if it were generated by a different mechanism*.

Outlier ≠ noise

No correlation

- **Noise**: random error or variance in a measured variable — uninteresting, removed in preprocessing
- **Outliers**: interesting because they violate the normal mechanism
- Noise inflates false alarms, so suppress it first

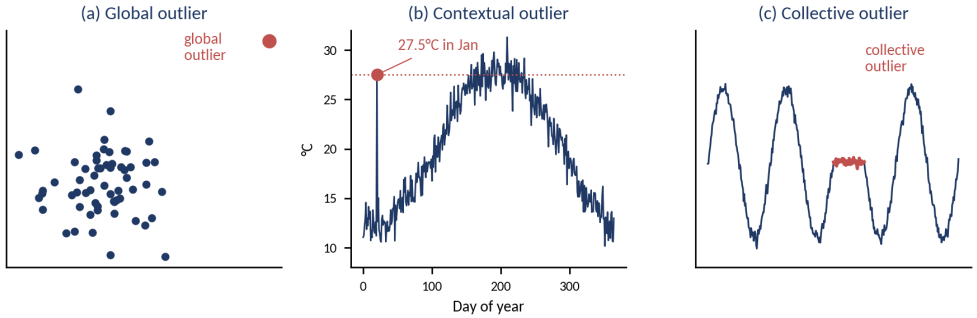
Correlated

Applications

- Credit-card and insurance fraud
- Network intrusion detection
- Medical diagnosis, patient monitoring
- Fault detection in industry and IoT
- Public safety; data cleaning

Global outlier ≠ noise

Three types of outliers



- **Global** (point anomaly): deviates from the whole data set
- **Contextual** (conditional): deviates within a context — **contextual attributes** (date, location) define the context; **behavioural attributes** (temperature) are judged
- **Collective**: a subset deviates as a whole, though members may be normal — needs relationships among objects (time, space, graph)

Challenges of outlier detection

1. **Modelling normal objects and outliers effectively** — the boundary is often a grey zone; enumerating all normal behaviour is impossible.
2. **Application-specific detection** — similarity measures and models depend on the domain: a small deviation is critical in clinical data, tolerable in marketing.
3. **Handling noise** — noise blurs the line between normal and abnormal and can hide true outliers.
4. **Understandability** — users want a justification and a *degree* of outlier-ness, not just a flag.



K = Boltzman Const.
T = T + 273
B = Band width

Methods by availability of labels

Supervised

Classification with expert-labelled normal and outlier examples.

Class imbalance: outliers are rare — use oversampling, cost-sensitive learning, recall-based evaluation.

Finds only outlier kinds it has seen.

Semi-supervised

A few labelled objects, usually normal, plus unlabelled data.

Labelled normal objects and close unlabelled neighbours train a model of normality; misfits are outliers.

Unsupervised

No labels. Assumes normal objects are **clustered**, outliers far from clusters.

Fails if normal data lack patterns or if outliers form their own cluster (coordinated intrusions).

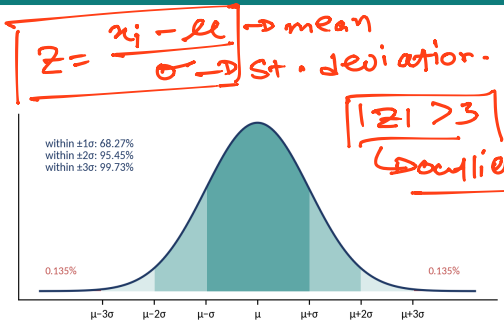
Methods by assumption about the data

Family	Core assumption	Typical methods
Statistical	Normal objects come from a statistical model; outliers lie in low-probability regions	3σ, Grubbs, Mahalanobis, mixtures, histograms, KDE
Proximity-based	An outlier's proximity to its neighbours differs markedly from that of most objects	DB(r, π), k-NN distance, LOF
Clustering-based	Normal objects belong to large dense clusters; outliers to small/sparse clusters or none	DBSCAN noise, distance to centroid, CBLOF
Classification-based	A classifier can learn a boundary around normal data	One-class SVM, semi-supervised classifiers

Univariate parametric detection

Assume $x \sim N(\mu, \sigma^2)$; estimate μ, σ by $\bar{x}, s$.

$$z_i = \frac{x_i - \bar{x}}{s}, \qquad \text{flag if } |z_i| > 3$$



- 99.73% of a normal distribution lies within $\mu \pm 3\sigma$
- **Boxplot rule** (distribution-free): beyond $Q_1 - 1.5\,IQR$ or $Q_3 + 1.5\,IQR$ — about $\mu \pm 2.7\sigma$ for normal data

Grubbs' test (maximum normed residual): outlier at level α if

$$G = \frac{\max_i |x_i - \bar{x}|}{s} > \frac{N-1}{\sqrt{N}} \sqrt{\frac{t^2_{\alpha/(2N),\,N-2}}{N - 2 + t^2_{\alpha/(2N),\,N-2}}}$$

Worked Example 1 — 3σ rule vs Grubbs' test

Ten temperature readings (°C): 24.1, 24.8, 25.0, 25.3, 25.6, 25.9, 26.2, 26.4, 27.0, 31.5

$\bar{x} = 26.18,\ s = 2.050,\ z(31.5) = (31.5 - 26.18)/2.050 = \mathbf{2.59}$

3σ rule

$|z| = 2.59 < 3 \Rightarrow$ not flagged

Grubbs ($\alpha = 0.05$, $N = 10$)

$t_{0.0025,\,8} = 3.833 \Rightarrow$ critical value $= \mathbf{2.290}$

$G = 2.59 > 2.29 \Rightarrow$ **outlier**

Masking The outlier inflates s. Without 31.5: $\bar{x} = 25.59$, $s = 0.894$, and 31.5 would be **6.6** standard deviations away. Prefer robust estimates (median, MAD, robust covariance) and formal tests.

Multivariate detection: Mahalanobis distance

An object can be normal on each attribute yet unusual in combination (tall but very light).

$$\text{MDist}^2(\mathbf{o}, \bar{\mathbf{o}}) = (\mathbf{o} - \bar{\mathbf{o}})^\top \mathbf{S}^{-1} (\mathbf{o} - \bar{\mathbf{o}}) \sim \chi^2_d \text{ for multivariate normal data}$$

- Accounts for variances **and correlations**; threshold at $\chi^2_d(0.95)$ or $\chi^2_d(0.99)$
- Or reduce to univariate: compute MDist for all objects, then apply Grubbs' test
- Cheaper χ^2 **statistic** treats dimensions independently — misleading when attributes are correlated:

$$\chi^2 = \sum_{i=1}^{d} \frac{(o_i - E_i)^2}{E_i}, \quad E_i = \text{mean of dimension } i$$

Worked Example 2 — Mahalanobis vs Euclidean

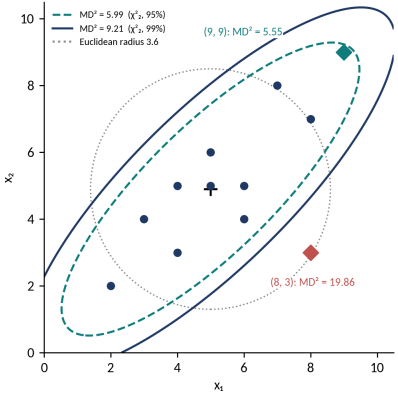
Ten objects; mean $(5, 4.9)$, $\mathbf{S} = \begin{bmatrix} 3.333 & 2.667 \\ 2.667 & 3.211 \end{bmatrix}$

($r \approx 0.81$)

Object	Euclid.	MDist^2	χ^2 stat.
$p = (8, 3)$	3.55	**19.86**	2.54
$q = (9, 9)$	5.73	5.55	6.63

$\chi^2_2(0.95) = 5.99,\ \chi^2_2(0.99) = 9.21$

$\Rightarrow$ *p* **is an outlier** at 1%; *q* is not at 5% — the reverse of the Euclidean ranking.



q lies along the correlation axis; *p* runs against it. With $n = 10$ the χ^2 approximation is rough — use a robust covariance (MCD) in practice.

Mixture of parametric distributions

$$\Pr(\mathbf{o} \mid \Theta) = \sum_{j=1}^{m} \pi_j f_j(\mathbf{o} \mid \theta_j)$$

Learn with EM; low-likelihood objects are outliers. Avoids assigning high probability to the empty region between sub-populations.

Non-parametric

- **Histogram**: score $\propto$ 1 / frequency of the object's bin. Bin width: too small $\Rightarrow$ false positives; too large $\Rightarrow$ false negatives
- **Kernel density estimation**:

$$\hat{f}_h(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{\mathbf{x} - \mathbf{x}_i}{h}\right)$$

Key idea Statistical methods justify flags probabilistically — if the model is right. Weaknesses: mis-specification, masking, density estimation in high dimensions.

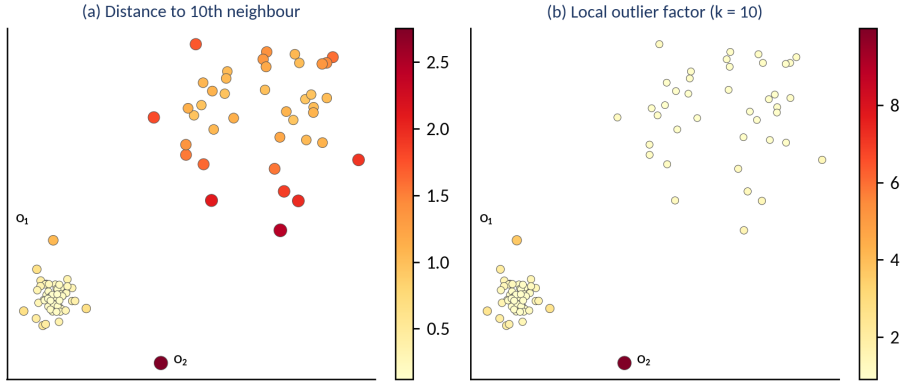
Distance-based outliers: DB(r, π)

$$\mathbf{o} \text{ is a DB}(r,\pi)\text{-outlier} \iff \frac{\left|\{\mathbf{o}' \in D \setminus \{\mathbf{o}\} : \ \mathrm{dist}(\mathbf{o},\mathbf{o}') \le r\}\right|}{|D|-1} \le \pi$$

- Equivalently: the distance to the k-th nearest neighbour, $k = \lceil \pi(|D|-1) \rceil$, exceeds r
- **Nested loop**: $O(n^2)$, but the inner loop stops once enough neighbours are found — normal objects in dense regions exit early
- **Grid-based** (CELL): cells of side $r/(2\sqrt{d})$; whole cells declared normal or outlying by counting neighbouring cells — efficient in low dimensions
- Index structures (R-tree, k-d tree) accelerate neighbour search

Key idea One **global** radius $r \Rightarrow$ trouble when the data contain regions of different density.

Why global distance fails: varying density



- (a) o_1's 10-NN distance (≈ 1.06) is *smaller* than that of many normal objects in the sparse cluster (up to ≈ 2.45) — no single radius works
- (b) LOF compares local densities: o_1 scores ≈ 3.7, while no sparse-cluster object exceeds ≈ 1.5

Local outlier factor (LOF)

$\text{dist}_k(\mathbf{o})$: distance to the k-th nearest neighbour; $N_k(\mathbf{o})$: objects within that distance (may exceed k with ties).

$$\text{reachdist}_k(\mathbf{o}, \mathbf{o}') = \max\{\text{dist}_k(\mathbf{o}'),\ \text{dist}(\mathbf{o}, \mathbf{o}')\}$$

$$\text{lrd}_k(\mathbf{o}) = \frac{|N_k(\mathbf{o})|}{\sum_{\mathbf{o}' \in N_k(\mathbf{o})} \text{reachdist}_k(\mathbf{o}, \mathbf{o}')}$$

$$\text{LOF}_k(\mathbf{o}) = \frac{1}{|N_k(\mathbf{o})|} \sum_{\mathbf{o}' \in N_k(\mathbf{o})} \frac{\text{lrd}_k(\mathbf{o}')}{\text{lrd}_k(\mathbf{o})}$$

- Reachability distance smooths statistical fluctuation for very close neighbours
- LOF ≈ 1: density like its neighbours (inside a cluster); LOF $\gg 1$: sparser than its neighbours $\Rightarrow$ outlier

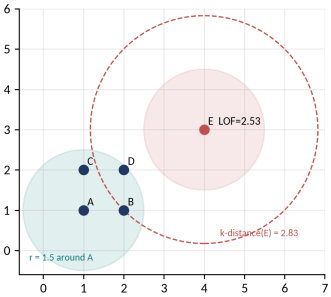
Worked Example 3 — DB(r, π) and LOF by hand

$A = (1,1)$, $B = (2,1)$, $C = (1,2)$, $D = (2,2)$, $E = (4,3)$

DB$(1.5,\ 0.25)$: A–D each have 3 of 4 others within 1.5 (0.75); E's nearest is $\sqrt{5} = 2.24$ away (0) $\Rightarrow$ E **is an outlier**

LOF, $k = 2$

- $\text{dist}_2(A..D) = 1;\ \text{dist}_2(E) = \sqrt{8} = 2.828,$ $N_2(E) = \{D, B\}$
- $\text{lrd}(A) = 2/(1+1) =$ **1** (same for B, C, D)
- $\text{lrd}(E) = 2/(\max\{1, 2.236\} + \max\{1, 2.828\}) =$ $2/5.064 =$ **0.395**
- $\text{LOF}(A..D) = 1;\ \text{LOF}(E) = 1/0.395 =$ **2.53**



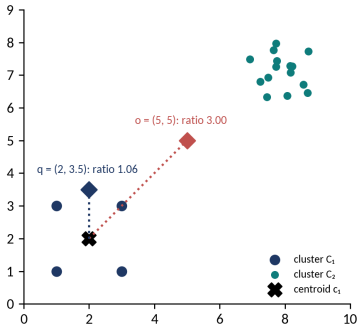
scikit-learn gives the same values

Three clustering-based criteria

1. **Belongs to no cluster?** DBSCAN labels non-core, non-reachable objects as **noise**
2. **Far from its closest cluster?** Distance to centroid, normalized by the cluster's spread:

$$\text{score}(\mathbf{o}) = \frac{\text{dist}(\mathbf{o}, \mathbf{c_o})}{\frac{1}{|C_\mathbf{o}|}\sum_{\mathbf{o}' \in C_\mathbf{o}} \text{dist}(\mathbf{o}', \mathbf{c_o})}$$

3. **Part of a small or sparse cluster? CBLOF**: size × distance to own centre (large clusters) or to nearest large cluster (small clusters)



Worked Example 4 — distance-to-centroid ratio

Cluster $C_1 = \{(1,1),(1,3),(3,1),(3,3)\}$, centroid $\mathbf{c}_1 = (2,2)$

Each member lies $\sqrt{2} \approx 1.414$ from $\mathbf{c}_1 \Rightarrow$ average member distance $= 1.414$

$o = (5,5)$: dist $= \sqrt{18} = 4.243 \Rightarrow$ ratio $= 4.243/1.414 = \mathbf{3.00}$ (strong candidate)

$q = (2,3.5)$: dist $= 1.5 \Rightarrow$ ratio $= 1.5/1.414 = \mathbf{1.06}$ (consistent with cluster)

Strengths

- Unsupervised; many data types
- Fast detection once clustered — compare against few clusters

Weaknesses

- Depends on clustering method and parameters
- Clustering aims to find clusters, not outliers; costly on large data

One-class models

- Two-class classifiers struggle: the outlier class is tiny and heterogeneous; new outliers may look unlike training outliers
- **One-class model**: learn only the normal class; anything outside the boundary is an outlier — can detect *new* kinds of outliers

One-class SVM: $\min_{\mathbf{w},\boldsymbol{\xi},\rho} \frac{1}{2}\|\mathbf{w}\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho \quad \text{s.t. } \mathbf{w}^\top \phi(\mathbf{x}_i) \geq \rho - \xi_i, \;\; \xi_i \geq 0$

- $\nu \in (0, 1]$ upper-bounds the fraction of training objects treated as outliers; kernel ϕ gives flexible boundaries
- **Semi-supervised**: cluster all data; clusters containing labelled normal objects are normal; objects far from them are outliers
- Fast at test time; quality depends on representative training data

Mining contextual outliers

Transform to conventional detection

When contexts are clearly identifiable: partition by contextual attributes and detect within each context.

Example: compare a customer's spending only with customers of the same age group and postal area; merge similar contexts if groups are small.

Model normal behaviour w.r.t. context

When contexts cannot be partitioned (time series): predict expected behaviour from contextual attributes (regression, time-series or Markov model).

Example: the model expects ≈ 12°C on a January day; an observed 27.5°C has a large residual.

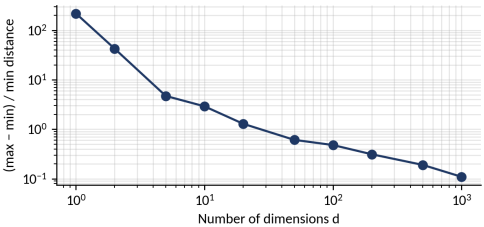
Mining collective outliers

- Require explicit or implicit **structure**: sequences, spatial neighbourhoods, graph links
- **Explicit structure**: define units (sub-sequences, regions, sub-graphs), compute unit-level features, apply a conventional detector
  - e.g. each 10-minute sensor window $\rightarrow$ one object described by mean, variance and spectral features
- **Implicit structure** (coordinated user behaviour): search for subsets with unusual joint behaviour — e.g. transactions co-occurring far more often than expected; much harder

Key idea Collective outliers often signal coordinated activity: denial-of-service bursts, fraud rings, sensor failures.

The curse of dimensionality for outliers

- Data become sparse; **distances concentrate** — nearest and farthest neighbours become almost equally far
- 1 attribute: farthest $\approx$ hundreds of times the nearest; 1,000 attributes: only $\approx 11\%$ farther
- An object may be outlying only in a small **subspace**, diluted by irrelevant dimensions
- Results must also be **interpretable**: which attributes make it an outlier?



500 uniform points; relative contrast vs dimensionality

Three lines of attack

1. **Extend conventional methods** — *HilOut*: rank by the *sum* of k-NN distances, approximated with space-filling curves; or reduce dimensionality first (outliers may hide in discarded directions)
2. **Find outliers in subspaces** — grid with φ equi-depth ranges per dimension ($f = 1/\varphi$); a k-d cell should hold about Nf^k objects. **Sparsity coefficient**:

$$S(C) = \frac{n(C) - Nf^k}{\sqrt{Nf^k(1 - f^k)}}$$ strongly negative $\Rightarrow$ objects in C are outliers

3. **Model high-dimensional outliers** — **ABOD**: from an outlier, other points lie in a narrow cone, so angles vary little

$$\mathrm{ABOF}(\mathbf{o}) = \mathrm{VAR}_{\mathbf{x},\mathbf{y} \in D} \frac{\langle \overrightarrow{\mathbf{ox}}, \overrightarrow{\mathbf{oy}} \rangle}{\|\overrightarrow{\mathbf{ox}}\|^2 \, \|\overrightarrow{\mathbf{oy}}\|^2}$$ small $\Rightarrow$ outlier; exact cost $O(n^3)$

Choosing a detector

Method	Strengths	Weaknesses
3σ / Grubbs	Simple; probabilistic justification	Assumes normality; masking; univariate
Mahalanobis + χ^2	Captures correlation; principled threshold	Elliptical data; needs robust covariance
Histogram / KDE	Few assumptions; explainable	Bin width / bandwidth sensitive; poor in high dimension
DB(r, π)	Intuitive; distribution-free	Global threshold; varying density; parameters
LOF	Handles varying density; graded score	$O(n^2)$ naïvely; sensitive to k
Clustering-based	Fast once clustered; finds small clusters	Depends on clustering quality
One-class SVM	Flexible kernel boundary; new outlier types	Needs representative normal data; tuning ν
ABOD / subspace	Built for high dimensions; explainable subspaces	Computationally expensive

Evaluation Use precision, recall and ROC/PR AUC — not accuracy. Flagging nothing gives 99% accuracy when 1% are outliers.

In practice: scikit-learn

Four detectors on contaminated synthetic data

```
from sklearn.covariance import EllipticEnvelope
from sklearn.neighbors import LocalOutlierFactor
from sklearn.svm import OneClassSVM
from sklearn.ensemble import IsolationForest

detectors = {
    "Mahalanobis (robust)": EllipticEnvelope(contamination=0.05),
    "LOF (k = 20)":         LocalOutlierFactor(n_neighbors=20, contamination=0.05),
    "One-class SVM":        OneClassSVM(nu=0.05, gamma="scale"),
    "Isolation forest":     IsolationForest(contamination=0.05),
}
for name, det in detectors.items():
    labels = det.fit_predict(X)          # +1 = normal, -1 = outlier
```

Isolation forest is a later ensemble method (not in the textbook) that isolates outliers with random splits — now a common baseline.

Check your understanding

1. Classify as global, contextual or collective: ten identical consecutive ECG samples; a 45°C boiler reading (normal 80–95°C running, 25–30°C idle).
2. For 12, 14, 15, 15, 16, 17, 18, 45: apply the 3σ rule and Grubbs' test ($\alpha = 0.05$, critical value 2.127).
3. Mean $(0, 0)$, $\mathbf{S} = \text{diag}(4, 1)$: compute MDist^2 of $(4, 1)$ and $(1, 3)$.
4. Compute the LOF of every object in $\{0, 1, 2, 3, 8\}$ with $k = 2$.
5. Why can a k-NN distance detector fail with 500 attributes?

Answers: (1) collective; contextual. (2) $z = 2.44 < 3$ not flagged; $G = 2.44 > 2.127$ outlier (masking). (3) 5.00 (not an outlier at 95%), 9.25 (outlier at 99%). (4) 1, 1, 1, 1, 3.67. (5) distance concentration and subspace dilution.

Further reading

- J. Han, M. Kamber and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011 — Chapter 12.
- M. M. Breunig, H.-P. Kriegel, R. T. Ng and J. Sander, “LOF: identifying density-based local outliers,” *Proc. ACM SIGMOD*, 2000.
- V. Chandola, A. Banerjee and V. Kumar, “Anomaly detection: a survey,” *ACM Computing Surveys*, 41(3), 2009.
- C. C. Aggarwal, *Outlier Analysis*, 2nd ed., Springer, 2017.

Thank you

Questions and discussion

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